Manuscript outline

Cell enlargement, polarity competition and migration persistence

Size-dependent polarity responses and prospective cellular measurements

23 October 2026 · Size, polarity response and heading persistence

Scientific argument and manuscript plan

Working hypothesis: WGD-associated enlargement changes how established polarity is organized and replaced; these changes may provide a physical basis for greater migration persistence. The theoretical contribution tests size under unchanged local kinetic parameters. Length differs between model cells to represent WGD-associated enlargement, rather than elongation during a turn. Existing migration observations supply empirical motivation; new paired WGD motility results retain a substantial, explicitly reserved contribution.

Figure 2 separates size-dependent initiation and cue-width responses from temporal class signatures: initiation-limited resetting, saturating activation with visible gap retention, and pulse-2 progress after loss of visible activity. The full Holmes width series shows finite replacement at all tested lengths for sufficiently broad cues; narrow-cue censoring cannot establish a growing penalty.

Cue placement separates sliding of the incumbent from takeover by a new region; enlargement slows both routes. Weak boundary susceptibility changes differently across models: terminal response at the largest size is 70% of the smallest-size response in Wave-pinning, 9% in Goryachev, 91% in Holmes and 99% in Spring. The deterministic boundary response predicts physical domain drift without fitting when shared translation is retained. Wave-pinning’s C60/C20 drift ratio at f = 0.001 is 0.494 predicted and 0.473 measured. Holmes and Spring show much weaker susceptibility changes. The independent-edge approximation is refuted by boundary co-motion; Goryachev’s drift prediction remains quantitatively inconsistent and does not explain its full heading persistence.

Enlargement extends heading memory and raises noise tolerance, while the size advantage weakens as input strengthens. The weak-input drift in Wave-pinning, Holmes and Spring is close to quadratic in input, connecting Figure 4 to the deterministic edge-response test in Figure 3. Direct event accounting distinguishes persistent secondary onsets from certifiable no-parent births. Certified takeover probability declines with size in each model at a common finite follow-up, but different certification fractions and pooled inputs limit population inference. Per-perimeter birth direction varies by

Manuscript outline · continued

model; it is not a universal consequence of enlargement. At native moment amplitude, eligible weak-input heading loss is allocated almost entirely to domain motion. As secondary domains appear, membership changes and the amplitude/shape/background remainder can grow; the pattern varies with model and size. This exact accounting remains descriptive, and the original unit-normalized point-region artefact is retained as a supplementary diagnostic.

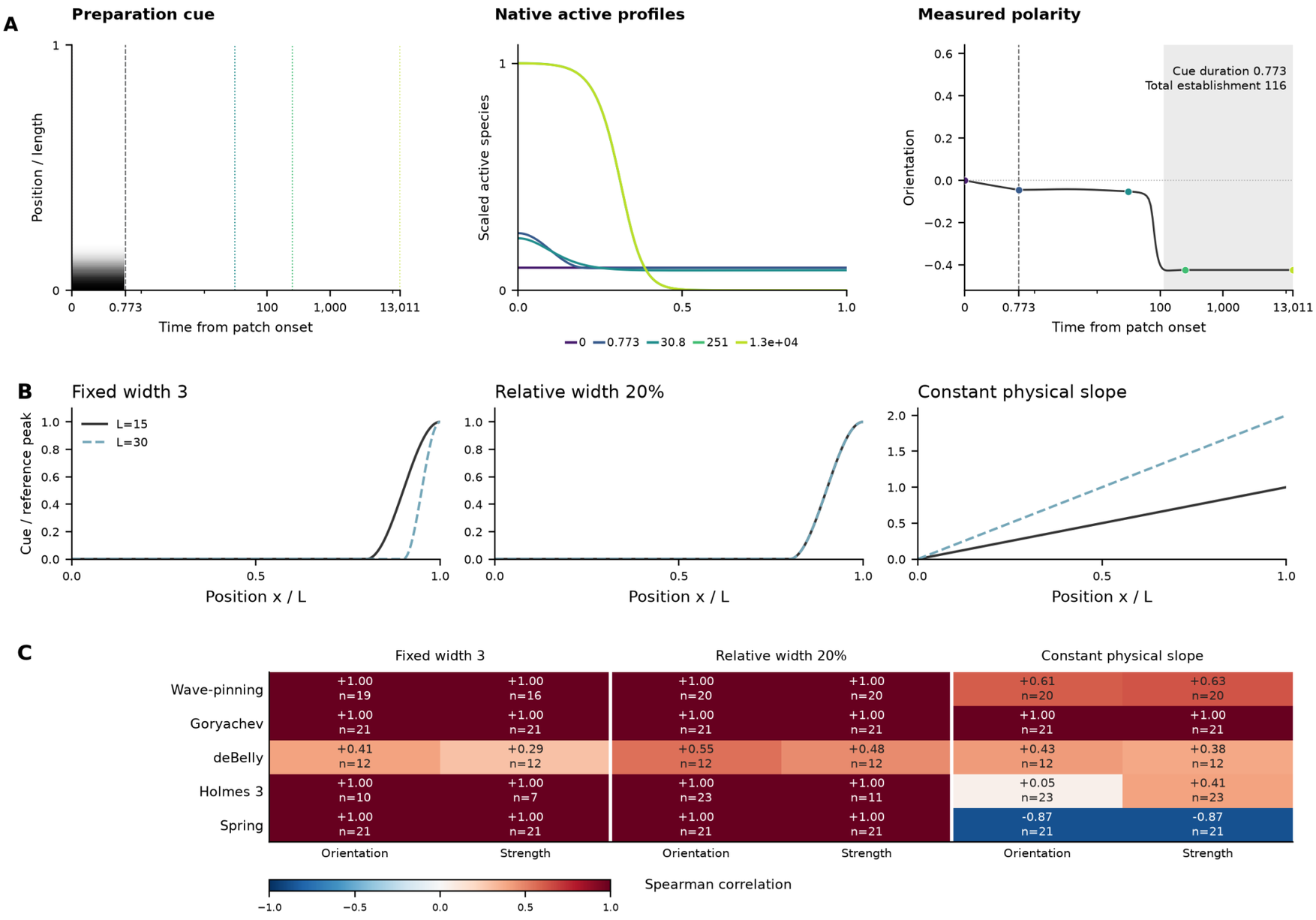
The prospective pilot table fixes model-linked cellular observables before the SUM-159 data arrive. Directions are conditional hypotheses; unavailable model measurements remain unavailable. Activity heading, observed migration summaries and conditional OU velocity-memory estimates remain separate evidence.

This is a working manuscript outline. Figures 1 and 5 and the reserved Figure 6 experimental section retain their preceding content. Activity polarity, modelled replacement, observed migration and conditional OU velocity memory remain distinct readouts; their biological connection is an experimental hypothesis.

Proposed main figures

- Figure 1. Enlargement changes the cue duration required for lasting polarity replacement
- Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures
- Figure 3. Enlargement slows both turning routes and selectively suppresses boundary susceptibility.
- Figure 4. Enlargement extends heading memory as fluctuating input changes domain dynamics
- Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation
- Figure 6. RESERVED: new WGD motility-assay results

Figure 1. Five models and three finite challenges



Current canonical comparison, 10 October Spearman revision; corrected independent-endpoint deBelly.

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement

Comparative theoretical foundation: five established models, three finite challenges, and two distinct scalar replacement gates.

Detailed legend

A, Example wave-pinning preparation from a homogeneous state using a finite left patch. The cue ends at 0.773 native time units, whereas the subsequent basal relaxation establishes the polarized profile later (reported establishment arrival about 116). Coloured profiles and orientation samples distinguish input duration from autonomous development. The grey late interval is verification, not additional stimulation. Profiles are scaled for display.

B, Three rightward finite challenges restart the identical qualifying nominal-amplitude, relative-20% basal checkpoint at each model and length: fixed physical support 3 (Spring 0.2 native length units), relative support 20%, and a linear ramp with constant physical slope. Patch peaks remain fixed within each model; a fixed-slope ramp has a larger peak in a longer domain. Goryachev patches use the independently selected, length-independent multiplier K=64 (peak 6.4), which makes this a response-permissive comparison rather than a calibrated equal biochemical dose across models.

C, Spearman rank correlation between length and the shortest tested passing pulse duration, separately for orientation replacement and strength replacement, across Wave-pinning, Goryachev, corrected deBelly, Holmes 3 and Spring. Each cell reports the correlation and included successful-length count. All audited scalar successes enter, including those failing the separate native-field tail criterion. At least four distinct successful lengths and nonconstant durations are required; censored and preparation-ineligible cases receive no invented time. The companion duration curves retain all lengths and field-status markers.

After each positive pulse, input returns to the native basal value for 10,000 native time units. Orientation requires all 201 final-500 samples to have scalar orientation above 0.05 and contrast at least 0.05, with orientation range below 0.005. Strength additionally requires every tail orientation to be within 1% of the absolute initial leftward scalar orientation; it does not compare individual pole heights or amounts. The separate native-field tail flag requires pointwise range and projected 500-unit drift within 0.1% of frozen field scales. The reported duration excludes preparation and recovery. It is the shortest passing tested duration, with sampled fail/pass boundaries refined to 1%; narrower unobserved intervals are not excluded.

deBelly uses independently commanded corrected endpoints, rather than the deprecated in-loop centered extension. Spring length and time retain native units and are not silently converted to micrometres or seconds. Model-specific axes and stimulus definitions prevent interpreting cross-model amplitudes or clocks as biologically calibrated equivalents.

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement · continued

Narrative points and conclusions

- Fixed-width and relative-width patches show perfectly increasing rank order among observed successes in Wave-pinning, Goryachev, Holmes 3 and Spring; enlargement often increases the finite stimulation required for a retained response.
- The response depends on the cue geometry and model. Spring has a negative constant-slope association (Spearman correlation -0.874 for both gates), while Holmes orientation under the constant-slope cue is nearly unassociated with length (0.047).
- Corrected deBelly curves are nonmonotone despite moderate positive rank correlations. The full duration curves and eligibility ranges are essential to the result.
- These are deterministic polarity-replacement experiments. They motivate, but do not measure, spontaneous polarity memory or migration persistence.

Proposed measurements and experimental follow-up

Mechanism-linked cellular follow-up would match local input support, amplitude and duration, track the original and challenged activity regions through stimulation and withdrawal, and distinguish orientation replacement from recovered activity strength. This validation is proposed and has no experimental results in this outline.

Scope, limitations and pending validation

Correlations are conditional on observed scalar success, rather than an unbiased summary of all possible sizes or a universal scaling law. Ineligible starts are distinct from responses unobserved by the finite horizon.

A retained scalar response can coexist with a changing native field. Only 13 of the 72 corrected deBelly scalar successes pass the separate native-field tail flag. Resolution checks from the earlier centered-deBelly campaign cannot qualify the corrected replacement thresholds.

Sources and reproducibility

Three-test protocol and Spearman rule

Correlation table

Corrected endpoint method

Current duration curves

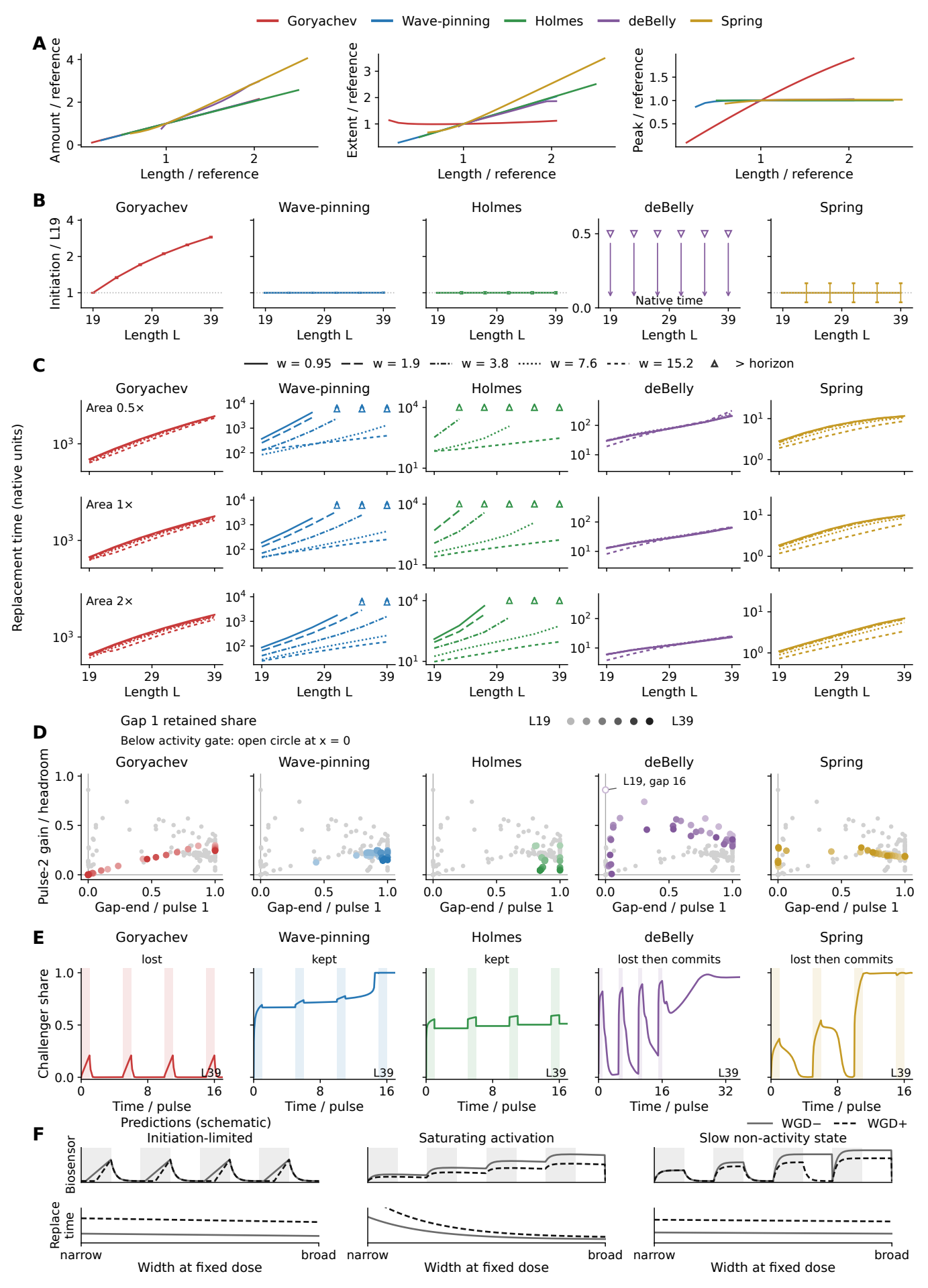

Goryachev
Wave-pinning
Holmes
deBelly
Spring
A
Amount / reference
Length / reference
Extent / reference
Peak / reference
B
Initiation / L19
Native time
Length L
C
w = 0.95
w = 1.9
w = 3.8
w = 7.6
w = 15.2
> horizon
Area 0.5×
Area 1×
Area 2×
Replacement time (native units)
D
Gap 1 retained share
Below activity gate: open circle at x = 0
L19
L39
L19, gap 16
Pulse-2 gain / headroom
Gap-end / pulse 1
E
lost
kept
lost then commits
Challenger share
Time / pulse
F
Predictions (schematic)
WGD−
WGD+
Initiation-limited
Saturating activation
Slow non-activity state
Biosensor
Replace time
narrow
broad
Width at fixed dose

Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures

Separate size-dependent initiation and cue-width responses from observable pulse-memory classes.

Detailed legend

Enlarged cells resist replacement through distinct initiation, cue-width and pulse-memory signatures.

A, Incumbent excess amount, extent and peak, normalized to each model's reference, across 97 retained states.

B, Initiation is the time from cue onset until maximum in-cue activity minus the profile minimum exceeds 10% of the model's L19 incumbent excess peak. Area 1, width 3.8; times are normalized to L19. All six deBelly upper bounds are 0.5 native units (downward arrows); first-update sampling leaves its size dependence unresolved.

C, Replacement time (first eligible challenger share ≥ 0.75) versus length, with a line per width and rows for areas 0.5, 1 and 2. Upward triangles/arrows are lower bounds: the cue never replaced within the horizon (6,000 Wave-pinning; 10,000 Goryachev/Holmes; 2,000 deBelly; 100 Spring, native units). Broad cues rescue Holmes across all lengths; narrow-cue censoring does not establish a growing penalty. Spring native widths equal the displayed widths divided by 38.

D, Gap-1 activity kept versus pulse-2 gain as a fraction of headroom (Eq. 1). Four equal pulses, total ON 1.5 times the model-length commitment calibration; gaps 0–16 pulse durations. Holmes L31–39 uses the L27 calibration. Other models are grey; lightness indicates L. Open circles at x = 0 code activity below the existing visibility gate. At the left wall, deBelly/Spring gain on pulse 2 despite no visible activity; Goryachev never does. The deBelly L19/gap16 first-gap endpoint was checked against native fields.

E, L39 gap4 challenger-share histories, labelled kept / lost / lost then commits; shading marks cue ON. deBelly includes full recovery; others show one post-train pulse duration. Continuous raw shares include below-gate values. deBelly's later timing and fields remain numerically sensitive (supplement).

F, Schematic biosensor predictions for the same pulse train and equal-dose width series in WGD− and enlarged WGD+ cells: delayed rises and resets; saturation with gap retention and strong width rescue; or gap collapse followed by commitment, with weaker width rescue. Predictions are prospective. Model kinetics are unchanged.

Narrative points and conclusions

- Enlarged cells resist replacement through distinct initiation, cue-width and pulse-memory signatures.

Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures · continued

- The full width series shows that Holmes can replace at every tested length when the cue is broad enough; narrow-cue censoring cannot establish a growing width-penalty ratio.
- Pulse-2 headroom gain separates low visible retention with residual progress from the initiation-limited reset response. Saturating activation retains visible activity through gaps.

Proposed measurements and experimental follow-up

Measure first in-cue biosensor rise, then vary width at fixed integrated dose.

Measure pulse-1, gap-1 and pulse-2 challenger shares under the same pulse train, followed by eventual commitment.

Scope, limitations and pending validation

Commanded input areas and native clocks are not calibrated across models.

deBelly initiation is unresolved within the first native update; later-gap timing and endpoint fields remain numerically sensitive.

D codes known below-gate activity at x=0; the exact raw ratio remains in the plotted-value CSV.

Holmes large-length trains have no target-length commitment calibration; visible retention differs from eventual replacement.

The three classes describe selected deterministic regimes; WGD biosensor predictions are prospective.

Model definitions and readouts

$$x = \frac{s_{\mathrm{gap\,1}}}{s_1}, \qquad y = \frac{s_2 - s_1}{1 - s_1} \tag{1}$$

s_1 and s_2 are challenger shares at pulse ends 1 and 2; s_gap1 is the first-gap-end share. An observed endpoint below the existing visibility gate is coded as zero visible share in D; its raw value remains in the CSV.

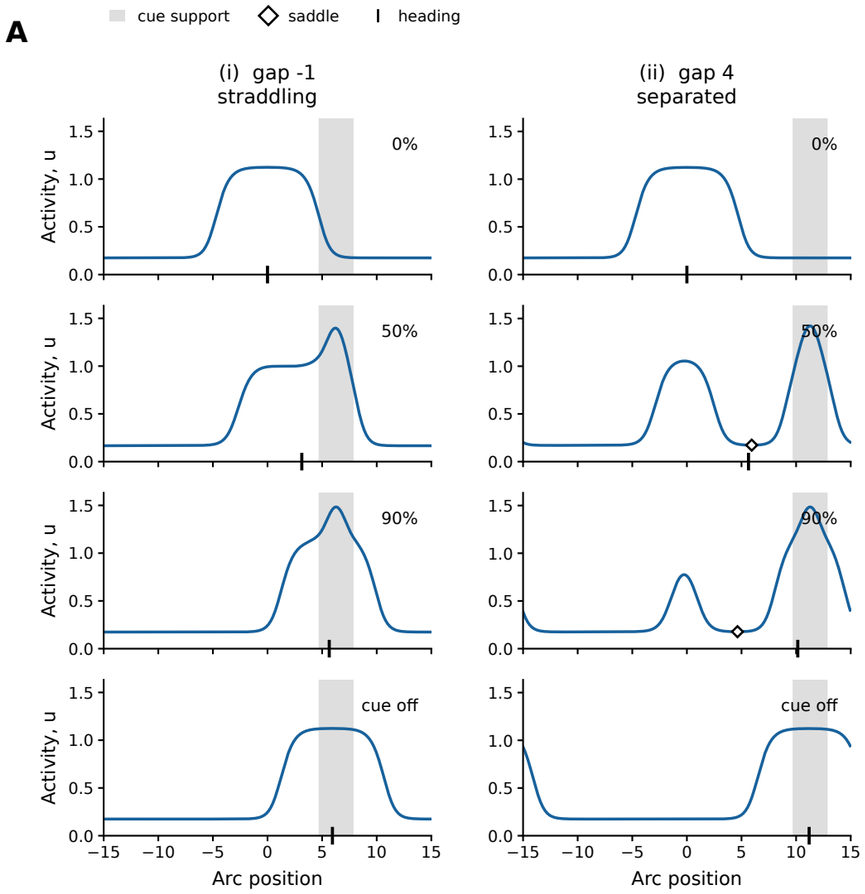
Sources and reproducibility

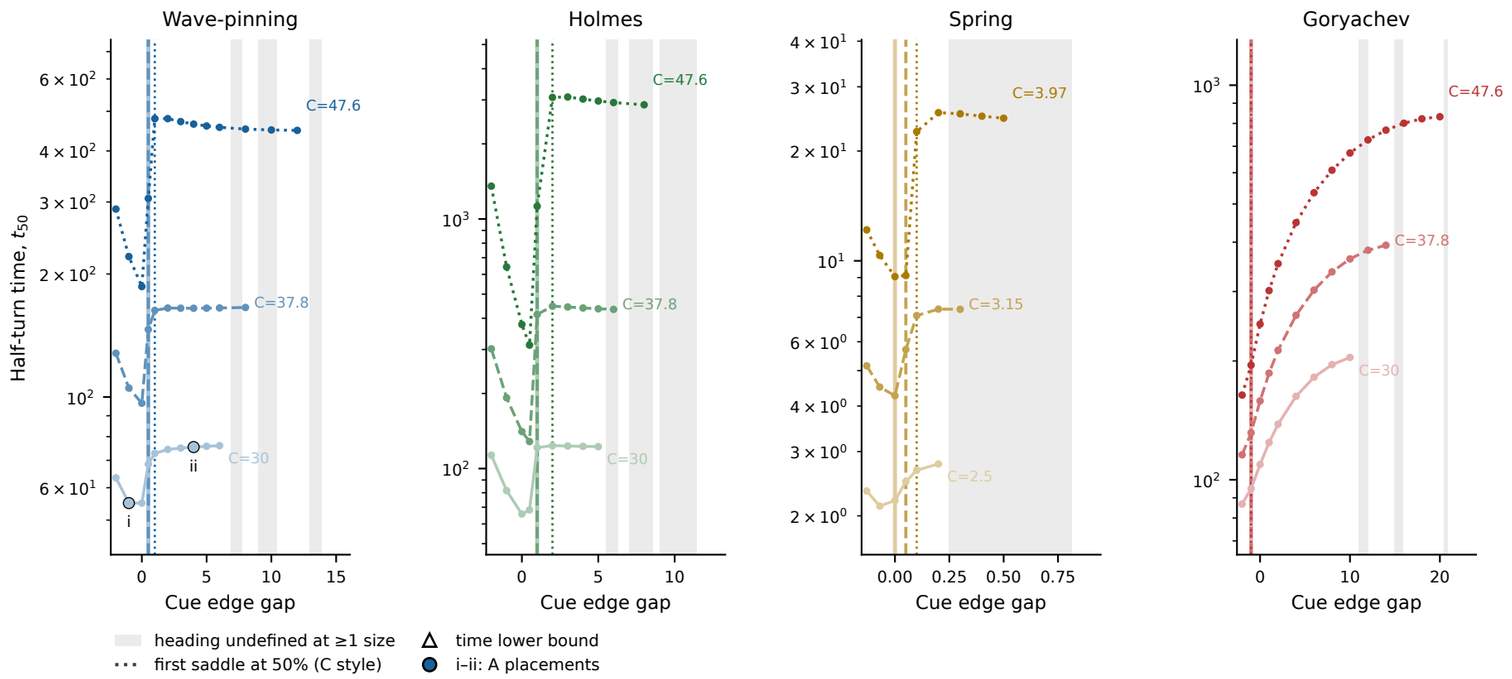
All 450 per-width replacement times

All 180 gap1/pulse2 coordinates and raw shares

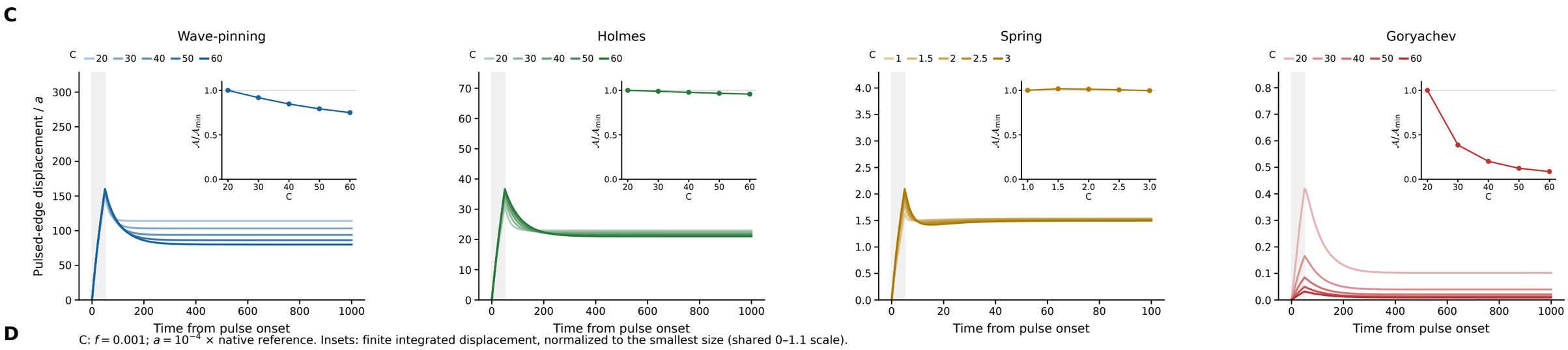
deBelly L19/gap16 native-phase audit

Preserved prior main panels and complete supplement



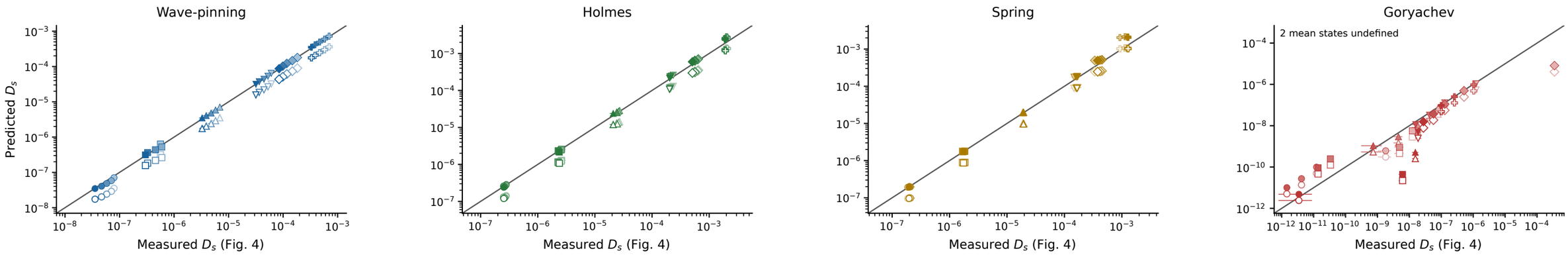


B: C = 30, 37.8, 47.6; Spring 2.5, 3.15, 3.97.
C–D: chemical C = 20–60; Spring 1–3. These are not matched size grids.



C: $f = 0.001$; $a = 10^{-4}$ × native reference. Insets: finite integrated displacement, normalized to the smallest size (shared 0–1.1 scale).

D



D: above, absolute diffusivity; below, size ratios at the same input. Lines: 1:1. Horizontal bars: whole-seed-bootstrap 95% CI.

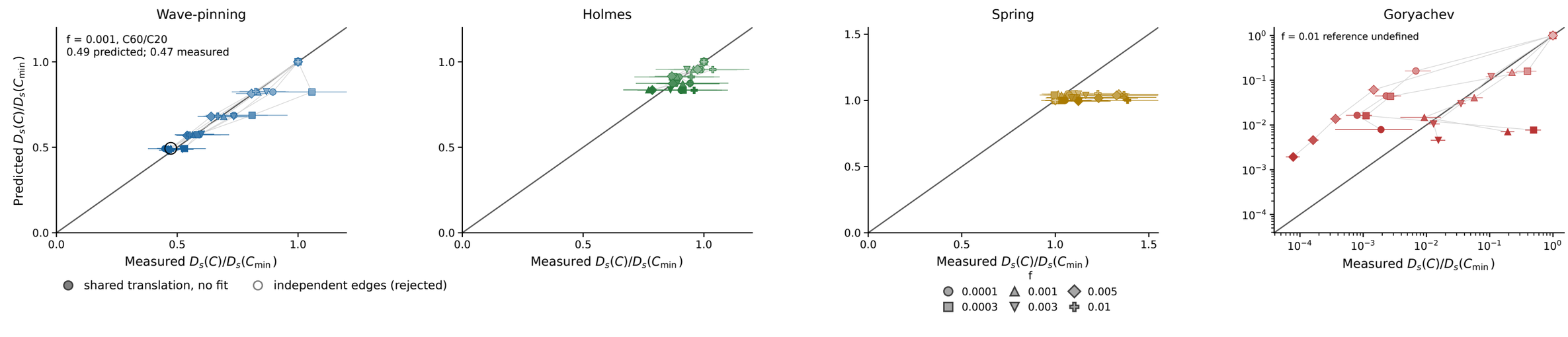


Figure 3. Enlargement slows both turning routes and selectively suppresses boundary susceptibility.

Size-dependent susceptibility links deterministic boundary perturbation to weak-input domain drift; cue position separates two slower turning routes in enlarged rings.

Detailed legend

A. Wave-pinning activity at C = 30 for a cue straddling the incumbent boundary (i, gap −1) or separated from it (ii, gap 4). Rows show 0%, 50% and 90% of cue-direction heading, then the relaxed cue-free profile. Grey: cue support; diamonds: inter-region saddles; ticks: heading.

B. Half-turn time versus cue-edge gap from the incumbent's 10%-contrast boundary. Increasing sizes use lighter solid, intermediate dashed and darker dotted curves; chemical C = 30, 37.8, 47.6, Spring 2.5, 3.15, 3.97. Vertical lines mark the first resolved two-region saddle at half-turn; grey bands mark undefined heading at ≥1 size; triangles are time lower bounds. Enlargement slows both turning routes. C–D use different size grids, so these panels do not establish a matched-size relation between strong-cue turning and weak-pulse response.

C. Control-subtracted displacement of the pulsed half-contrast boundary divided by peak input a, at f = 0.001. A Gaussian pulse has SD ℓ = 1.5 and duration τ = 50 (Spring 0.15 and 5); peak a = 10^{-4} of the native input reference. Shades denote chemical C = 20–60 or Spring resting C = 1–3. Insets show displacement integrated to 20τ, divided by pulse dose aτ, then normalized to the smallest size on a common 0–1.1 axis. Largest/smallest terminal gains are 0.701 Wave-pinning, 0.914 Holmes, 0.989 Spring and 0.0878 Goryachev. Both edges and half/double amplitudes are checked in the supplement.

D. Above: predicted versus Figure 4 physical tracked drift for f ≤ 0.01. Filled markers retain shared translation; open markers retain the rejected independent-edge candidate. Measured pulse gain sets response amplitude; the unchanged native linearization supplies spatial shape, and the exact input covariance supplies forcing statistics. Nothing is fitted to drift. Below: predicted versus measured $D_s(C)/D_s(Cmin)$ at the same f, removing the dominant input-strength scaling. Wave-pinning at f = 0.001 gives C60/C20 = 0.49 predicted and 0.47 measured. Lines: 1:1; bars: whole-seed-bootstrap 95% CI. Goryachev C20/C30 at f = 0.01 have no polarized mean-state boundary; the normalized prediction at that f also lacks its C20 reference.

Wave-pinning, Holmes and Spring have shared-translation prediction/observation medians 1.021, 1.080 and 1.069. Resolved quarter-window edge correlations are 0.980, 0.983 and 0.997 in the two exemplar seeds, refuting edge independence. Goryachev retains large quantitative disagreement. Tracked drift is conditional on trackable domains and is distinct from total heading decay.

Two cellular assays follow: vary photoactivation offset from straddling to separated placement and measure half-turn time; separately apply a weak brief boundary pulse and

Figure 3. Enlargement slows both turning routes and selectively suppresses boundary susceptibility. · continued

track both boundaries through withdrawal. The second assay measures susceptibility and common translation; it does not uniquely assign Figure 2 mechanism classes. Native units are not calibrated across models.

Narrative points and conclusions

- Cue placement separates sliding of the incumbent from takeover by a new region; enlargement slows both routes. Weak boundary susceptibility changes differently across models: terminal response at the largest size is 70% of the smallest-size response in Wave-pinning, 9% in Goryachev, 91% in Holmes and 99% in Spring.
- The deterministic boundary response predicts physical domain drift without fitting when shared translation is retained. Wave-pinning's C60/C20 drift ratio at f = 0.001 is 0.494 predicted and 0.473 measured. Holmes and Spring show much weaker susceptibility changes. The independent-edge approximation is refuted by boundary co-motion; Goryachev's drift prediction remains quantitatively inconsistent and does not explain its full heading persistence.

Proposed measurements and experimental follow-up

Vary photoactivation offset from straddling to separated placement; measure time to half-turn.

Apply a weak brief pulse across the leading-edge boundary; measure both boundary displacements, common translation and covariance. This readout complements rather than uniquely identifies the Figure 2 mechanism classes.

Scope, limitations and pending validation

Panel B and C–D use different size grids and perturbation strengths; a matched-size comparison is not established. One pulse does not identify spatial response shape; native linearization supplies that explicit assumption, while measured pulse gain supplies amplitude. Two float16 exemplar seeds support the covariance check, with short-lag precision limits. Goryachev's unresolved mean-state boundaries and quantitative drift disagreement remain visible. Physical domain drift is not total heading decay or empirical cell migration persistence.

Model definitions and readouts

$$D_s^{\mathrm{ind}} = \frac{D_L + D_R}{4}, \qquad D_s = \frac{D_L + D_R + 2D_{LR}}{4} \tag{3.1}$$

Open markers test independent edges; filled markers retain the untuned shared-translation covariance.

Figure 3. Enlargement slows both turning routes and selectively suppresses boundary susceptibility. · continued

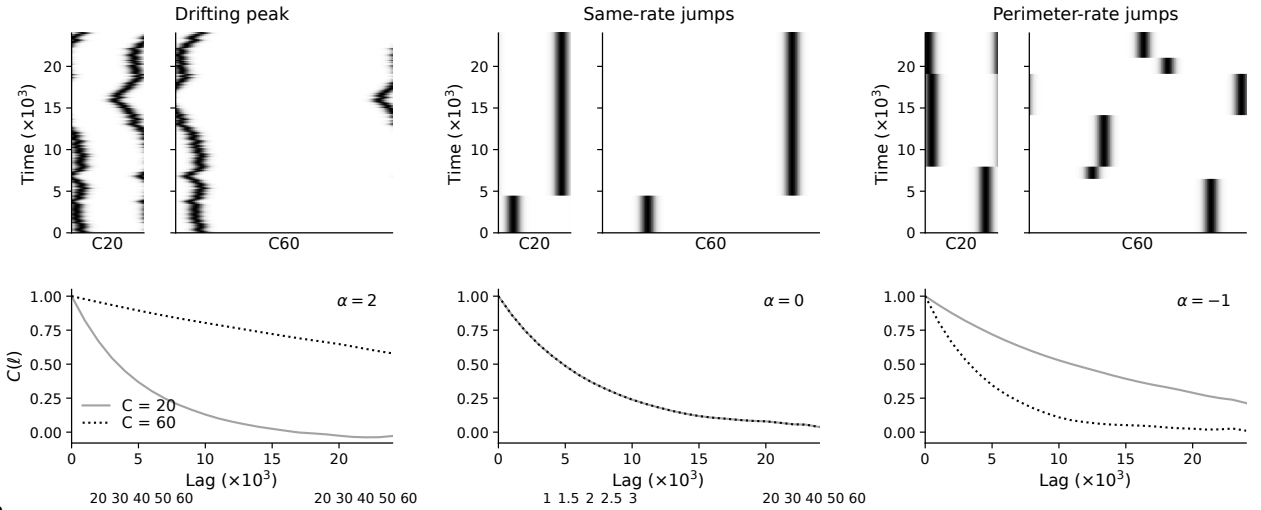
$$\widehat{g}_{e,j} = \frac{M_e}{Q_e} g_j, \qquad M_e = \frac{\Delta x_e(20\tau)}{a\tau}, \quad Q_e = \sum_j g_j h_{e,j} \tag{3.2}$$

Measured terminal displacement per pulse dose supplies amplitude; the unchanged native response supplies spatial shape. The covariance derivation is in the methods and supplement.

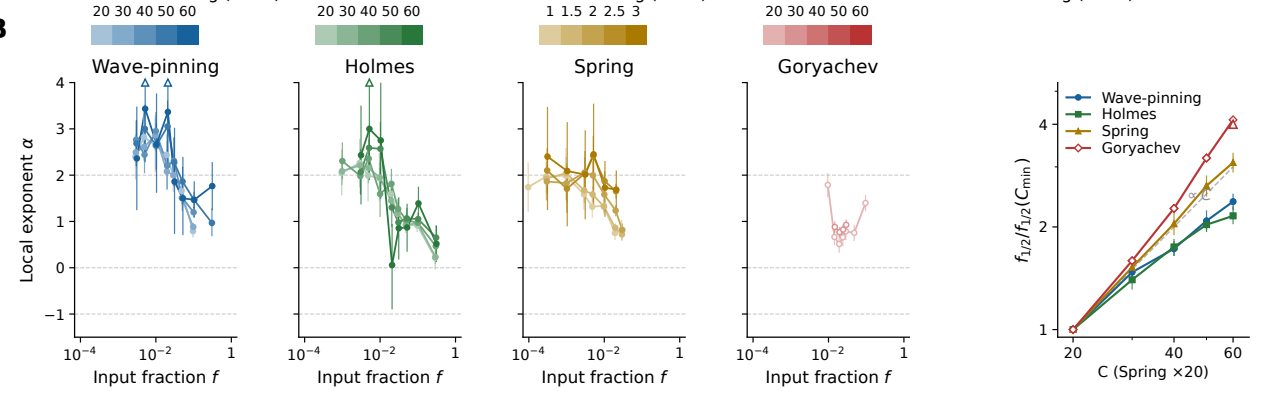
Sources and reproducibility

Physical diffusion prediction and observed drift
Size-normalized prediction and observed drift
Normalized integrated boundary response
Methods and covariance derivation
Both boundary assays and numerical checks
Reviewed main/supplement preservation

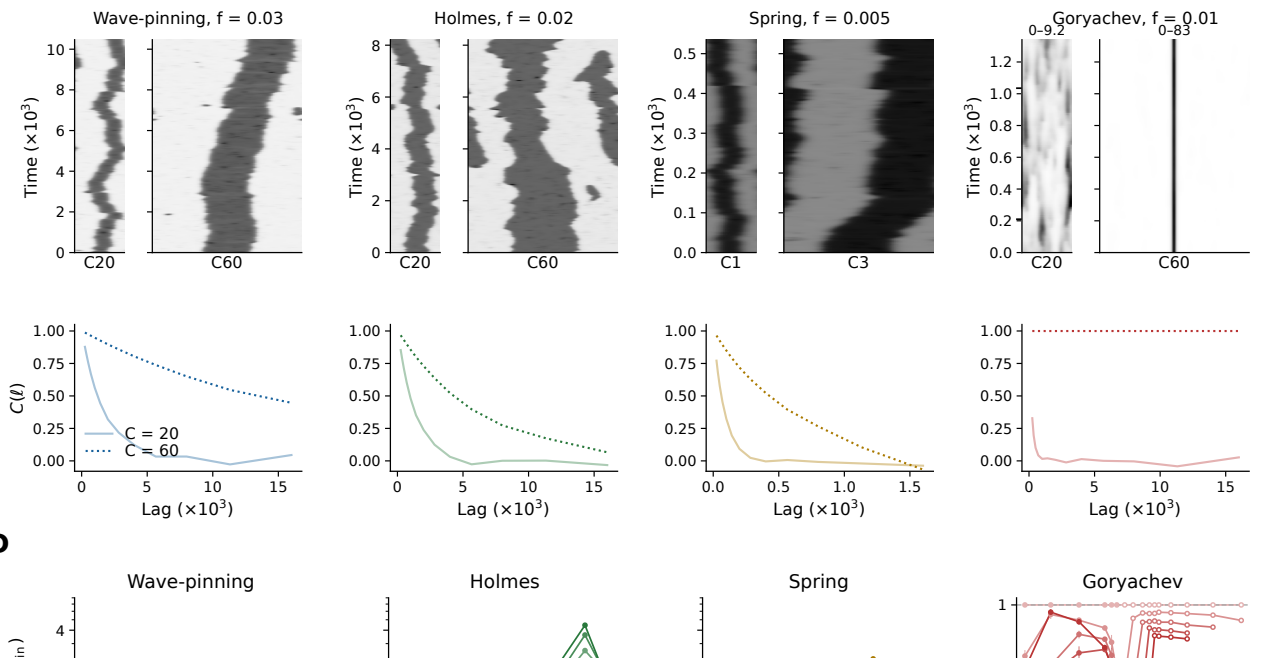
A


Drifting peak
Same-rate jumps
Perimeter-rate jumps
Time (×10³)
C20
C60
α = 2
α = 0
α = −1
C(ℓ)
C = 20
C = 60
Lag (×10³)

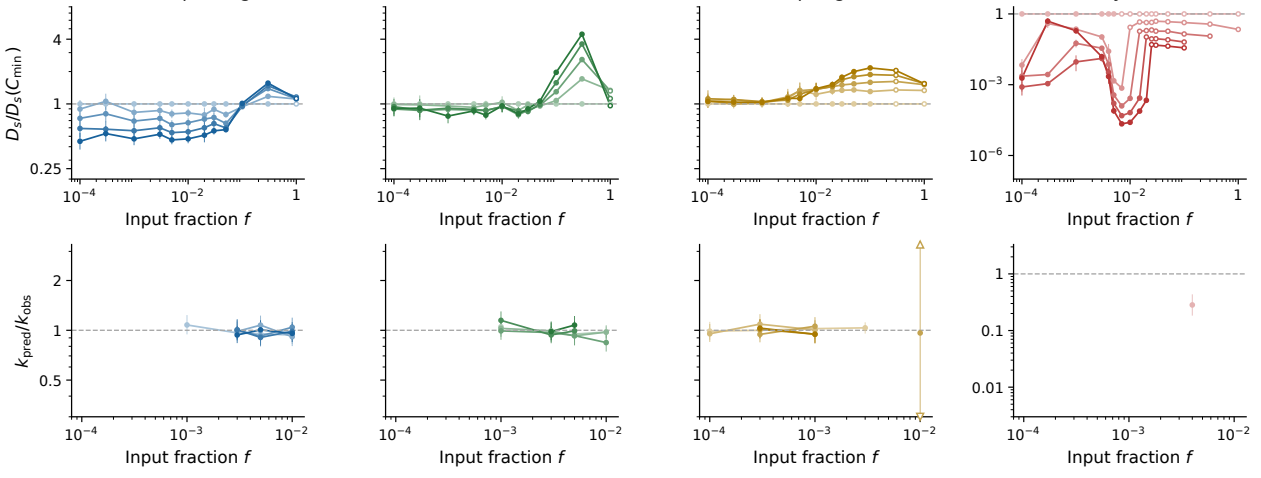
B


20 30 40 50 60
1 1.5 2 2.5 3
Wave-pinning
Holmes
Spring
Goryachev
Local exponent α
Input fraction f
Wave-pinning
Holmes
Spring
Goryachev
f1/2/f1/2(Cmin)
C (Spring ×20)

C


Wave-pinning, f = 0.03
Holmes, f = 0.02
Spring, f = 0.005
Goryachev, f = 0.01
0–9.2
0–83
Time (×10³)
C20
C60
C1
C3
C(ℓ)
C = 20
C = 60
Lag (×10³)

D


Wave-pinning
Holmes
Spring
Goryachev
Ds/Ds(Cmin)
kpred/kobs
Input fraction f

E

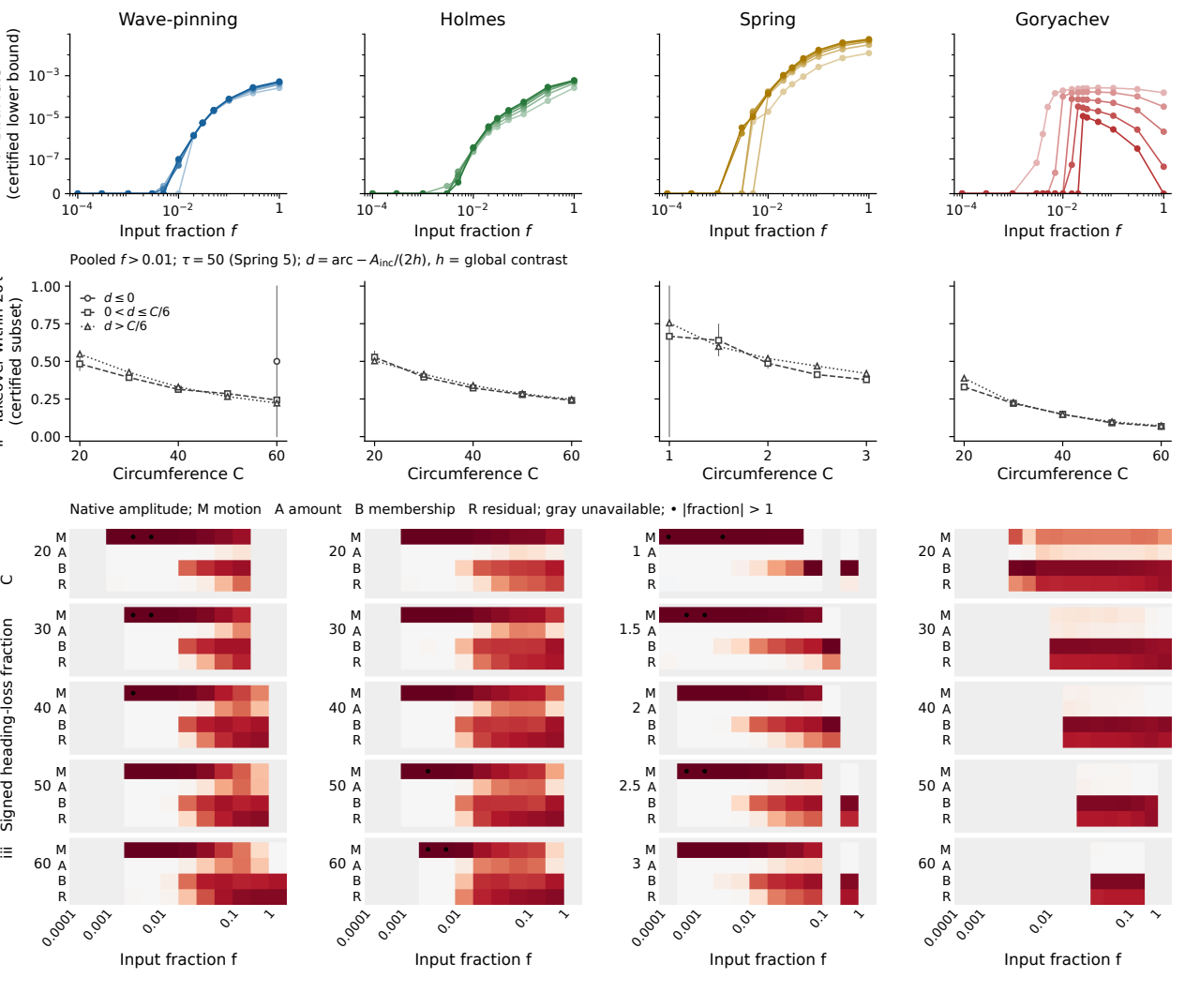

Wave-pinning
Holmes
Spring
Goryachev
i Birth rate (certified lower bound)
Input fraction f
Pooled f > 0.01; τ = 50 (Spring 5); d = arc − Ainc/(2h), h = global contrast
ii Takeover within 20τ (certified subset)
d ≤ 0
0 < d ≤ C/6
d > C/6
Circumference C
Native amplitude; M motion A amount B membership R residual; gray unavailable; • |fraction| > 1
iii Signed heading-loss fraction
C
M
A
B
R
Input fraction f

Figure 4. Enlargement extends heading memory as fluctuating input changes domain dynamics

Size-dependent heading persistence, edge drift and secondary-domain competition under fluctuating input.

Detailed legend

A. Drifting peaks, same-rate null relocations and perimeter-rate null relocations give size exponents 2, 0 and −1. Null kymograph widths follow circumference; time increases upward. Autocorrelations C(ℓ) use solid C20 and dotted C60 curves. The null relocations are illustrative whole-peak moves, distinct from measured dominant-region replacements and birth-to-takeover events.

B. Larger rings retain heading across stronger fluctuating input. Input fraction f sets the ensemble mean to f times the native reference (Wave-pinning 0.1; Holmes 1; Spring 4; Goryachev 6.4). Input is positive lognormal, mean × exp(σZ−σ²/2), with unit-variance Gaussian driving field Z and σ=1. The spatial parameter ℓ and OU correlation time τ are 1.5 and 50 in chemical models; Spring uses basal-physical ℓ=0.15 and τ=5, with physical ℓ scaling with its changing circumference. Local size exponents α and noise-tolerance folds use model hues and light-to-dark circumference ramps. Open symbols in the exponent and drift panels identify conditions flagged as forced by the frozen matched-mean/lower-noise controls; open diamonds in the fold panel identify Goryachev irrespective of forcing; vertical intervals are pointwise 95% whole-history bootstrap intervals. Boundary triangles retain off-scale values. 125/260 local neighborhoods resolve, including 11 Goryachev neighborhoods. Largest-size tolerance folds are Wave-pinning 2.373, Holmes 2.155, Spring 3.088 and Goryachev 4.122. $f_{1/2}$ is the first input crossing of C(ℓ)=0.5 at reference lag 2000 (Spring 200); Spring circumference is multiplied by 20 only in the fold panel.

C. Matched small- and large-ring kymographs are centred on their initial activity heading to the nearest spatial bin. Time increases upward; black outside ticks mark persistent dominant-region replacements. Activity scales are shared within Wave-pinning, Holmes and Spring; Goryachev uses 0–9.2 at C20 and 0–83 at C60. Ensemble C(ℓ) curves are below. A low-contrast initial heading may have weak biological meaning despite a defined numerical centre.

D. Physical within-tenure drift D_s is normalized by the smallest size at the same input. Unity gives the equal-physical-drift geometric expectation α=2. Weak-input log–log fits for f≤0.01 give exponents Wave-pinning 1.96–1.99; Holmes 1.94–1.95; Spring 1.87–1.93. The second row shows the existing event-based predicted/observed decay only where f≤0.01 and multiple regions occupy at most 5% of frames. 45/47 eligible point comparisons lie within 15% (Wave-pinning 16/16; Holmes 15/16; Spring 14/14; Goryachev 0/1). These point predictions assign the scored-event mean angular loss to unscored replacements; this remains an assumption. Goryachev uses separate logarithmic ordinate ranges: 10^{-7}–2 for normalized drift and 0.003–3.3 for decay ratio;

Figure 4. Enlargement extends heading memory as fluctuating input changes domain dynamics · continued

other models use 0.2–8 and 0.3–3.3. The actual mean circumference, $\bar{C}$, enters physical-to-angular conversion. Figure 3D independently predicts this measured drift from deterministic edge responses.

E. i, Certified no-parent secondary births per actual perimeter-time: new tracks persist at least five frames, have a preceding incumbent, and have conservative support enclosures disjoint from all previous regions. Rates are lower bounds; discarded masks prevent exact birth/split classification. ii, Observed incidence of five-frame global dominance by 20τ (1000 chemical units; 100 Spring), conditional on certified births with that follow-up. Inputs f>0.01 are pooled. Distance d subtracts estimated incumbent half-width (preceding full-basin amount divided by twice global contrast h) from the shortest centroid arc. Vertical intervals are 95% whole-history bootstrap intervals. A lower finite-horizon incidence can reflect delayed or less frequent takeover; lifetime success remains unidentified. Figure 2B–C motivates separating initiation from incumbent removal. iii, Native-amplitude-aware signed heading-loss allocation over the condition's analysis lag. M, shared-region motion; A, coexisting amount; B, track membership including unresolved splits/mergers; R, amplitude/shape/background remainder. Point-region Shapley changes are scaled by mean native endpoint moment amplitude, so pure rigid translation gives M=1, R=0. Components sum exactly to observed heading loss but are descriptive, convention-dependent shares. The symmetric-log color scale is linear within ±0.05 and spans ±1; dots indicate values outside ±1, gray unavailable pairs. The unit-normalized point-region diagnostic and its normalization artefact are preserved in the supplement; eligibility and equations are detailed in Methods.

Narrative points and conclusions

- Enlargement extends heading memory and raises noise tolerance, while the size advantage weakens as input strengthens. The weak-input drift in Wave-pinning, Holmes and Spring is close to quadratic in input, connecting Figure 4 to the deterministic edge-response test in Figure 3.
- Direct event accounting distinguishes persistent secondary onsets from certifiable no-parent births. Certified takeover probability declines with size in each model at a common finite follow-up, but different certification fractions and pooled inputs limit population inference. Per-perimeter birth direction varies by model; it is not a universal consequence of enlargement.
- At native moment amplitude, eligible weak-input heading loss is allocated almost entirely to domain motion. As secondary domains appear, membership changes and the amplitude/shape/background remainder can grow; the pattern varies with model and size. This exact accounting remains descriptive, and the original unit-normalized point-region artefact is retained as a supplementary diagnostic.

Figure 4. Enlargement extends heading memory as fluctuating input changes domain dynamics · continued

Proposed measurements and experimental follow-up

Measure lateral-edge fluctuations, secondary-protrusion births per actual perimeter-time, finite-horizon takeover probability and leading-edge displacement after boundary-straddling photoactivation. Preserve separate WGD-associated cell-size, observed migration and simulated polarity-heading readouts.

Scope, limitations and pending validation

Exact no-overlap birth counts and split counts cannot be recovered for the full campaign because compact records discard native masks/parent IDs. Lower/upper birth bounds and conditional takeovers are reported instead. The global-contrast plateau estimate is not a recovered per-region plateau.

Native clocks are not calibrated between models. A takeover is five-frame global dominance within 20 noise correlation times, not a lifetime success probability. Full-history right censoring, short-followup exclusions and per-condition counts are retained in CSV. Inputs f>0.01 are pooled without reweighting; size cohorts can differ in their event-weighted input composition. A lower incidence by 20τ does not distinguish slower takeover from a lower lifetime probability. A fixed-cohort supplementary incidence curve extends observation to 80τ without extrapolating beyond it.

Heading attribution uses only eligible lag endpoint pairs with measured $|P|\geq 0.1$, finite tracked amounts/centroids and all nonzero hybrid totals. Goryachev at f=0.03 has pair coverage 0.666 at C20 but only 0.0109 at C60. Endpoint membership does not reveal events occurring and reversing inside the lag. This algebraic allocation depends on the chosen convention and includes normalized point-region approximation error in its residual.

Historical multiple-front-majority closure point predictions are conditional on scored angles representing unscored replacements: none of 33 point estimates is within 15%, but missing-angle bounds permit agreement within 15% in 24/33 and span unity in 18/33. This cannot generally distinguish omitted heading processes from unscored large-angle replacements. Goryachev’s 37 historical finite point ratios span 0.00575–0.344, with limited dominant-region exposure and angle scoring; missing-angle bounds span unity in 11/37. These diagnostics remain in the supplement and do not motivate the independent exact endpoint allocation.

Model definitions and readouts

Figure 4. Enlargement extends heading memory as fluctuating input changes domain dynamics · continued

$$Q_t = \frac{\sum_{i \in I_t} A_{i,t} e^{i\theta_{i,t}}}{\sum_{i \in I_t} A_{i,t}}, \qquad r_t = |P_t|, \quad \widetilde{Q}_t = r_t Q_t \tag{4.1}$$

Q is the normalized point-region moment; P=M₁/M₀ is measured from the native field. Its amplitude r prevents the unit-modulus single-region normalization artefact.

$$\Delta P_j = \bar{r}\,\Delta Q_j \ (j = \mathrm{M}, \mathrm{A}, \mathrm{B}), \qquad \Delta P_{\mathrm{R}} = \Delta P - \bar{r}\,\Delta Q \tag{4.2}$$

The three ΔQ components use the symmetric Shapley convention for motion, coexisting amount and membership; r̄ is mean endpoint native amplitude. R closes the native moment change and includes amplitude variation, shape and background. This is an accounting convention, not causal attribution.

$$L_j = \frac{\mathrm{Re}\left[\Delta P_j \overline{(H_1 - H_0)}\right]}{|P_0| + |P_1|}, \quad H_t = \frac{P_t}{|P_t|}, \quad \sum_j L_j = 1 - \cos(\theta_1 - \theta_0) \tag{4.3}$$

Pooled signed fractions are sums of L_j divided by total measured heading loss. They retain cancellation, can exceed one, and depend on the stated accounting convention; they are not causal shares.

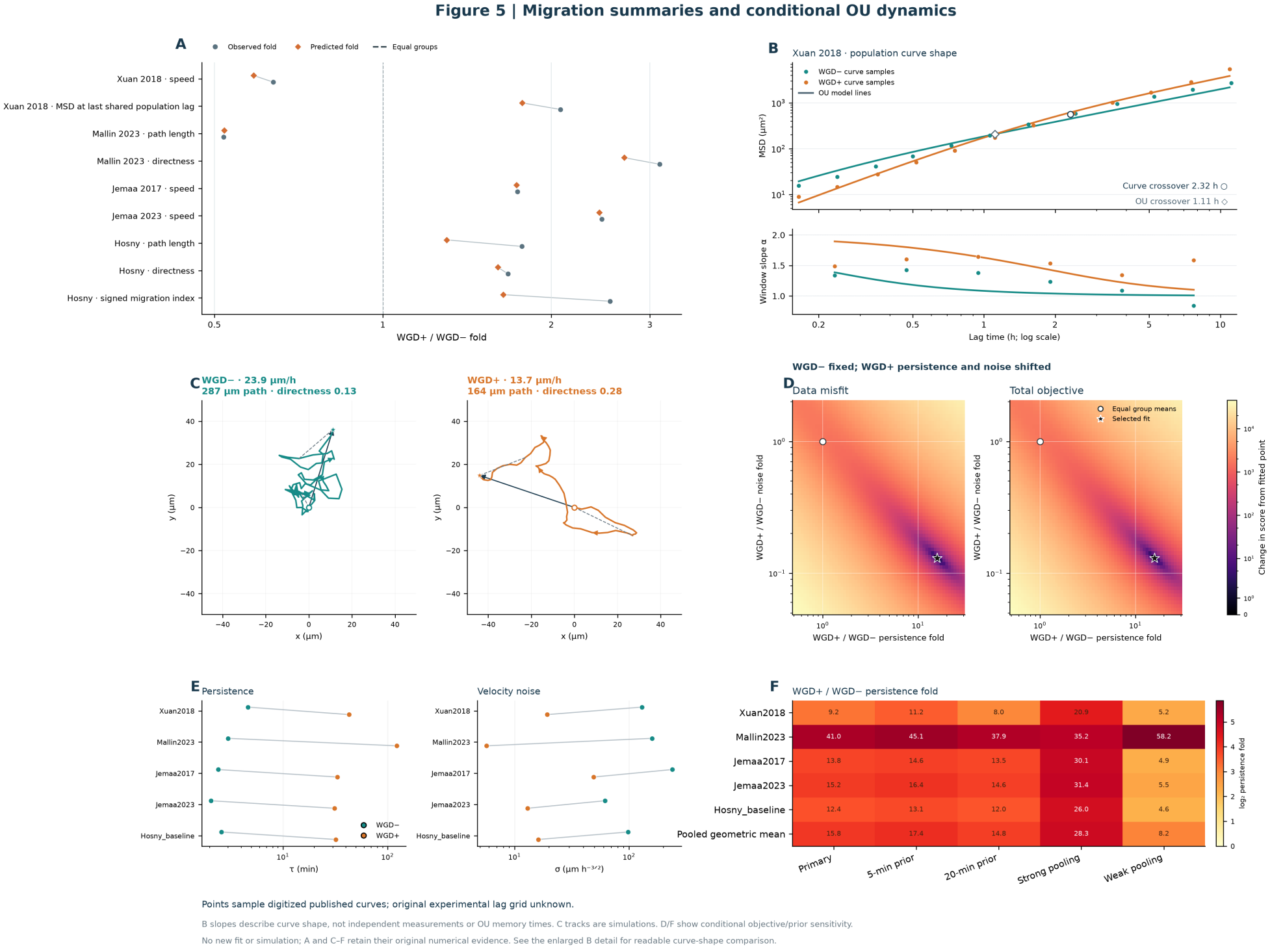
$$k(\ell) = -\frac{\ln C(\ell)}{\ell}, \qquad k \propto \bar{C}^{-\alpha} \tag{4.4}$$

C(ℓ) is activity-heading autocorrelation; C̄ is actual mean circumference. Endpoint eligibility uses |P|≥0.1. Drift closure is shown only at f≤0.01 and multiple-region fraction≤0.05.

Sources and reproducibility

Round5 complete campaign analysis
Scientific identity/enclosure validation
Birth and heading-loss conditions
Takeovers with common follow-up
Original campaign manifest
Native-amplitude heading accounting
Native-amplitude condition metrics

Figure 5. Migration observations and conditional OU interpretation · image 1 of 2



Reassembled Figure 5: original A and C–F numerical evidence, revised MSD/curve-shape panel B.

Figure 5. Migration observations and conditional OU interpretation · image 2 of 2

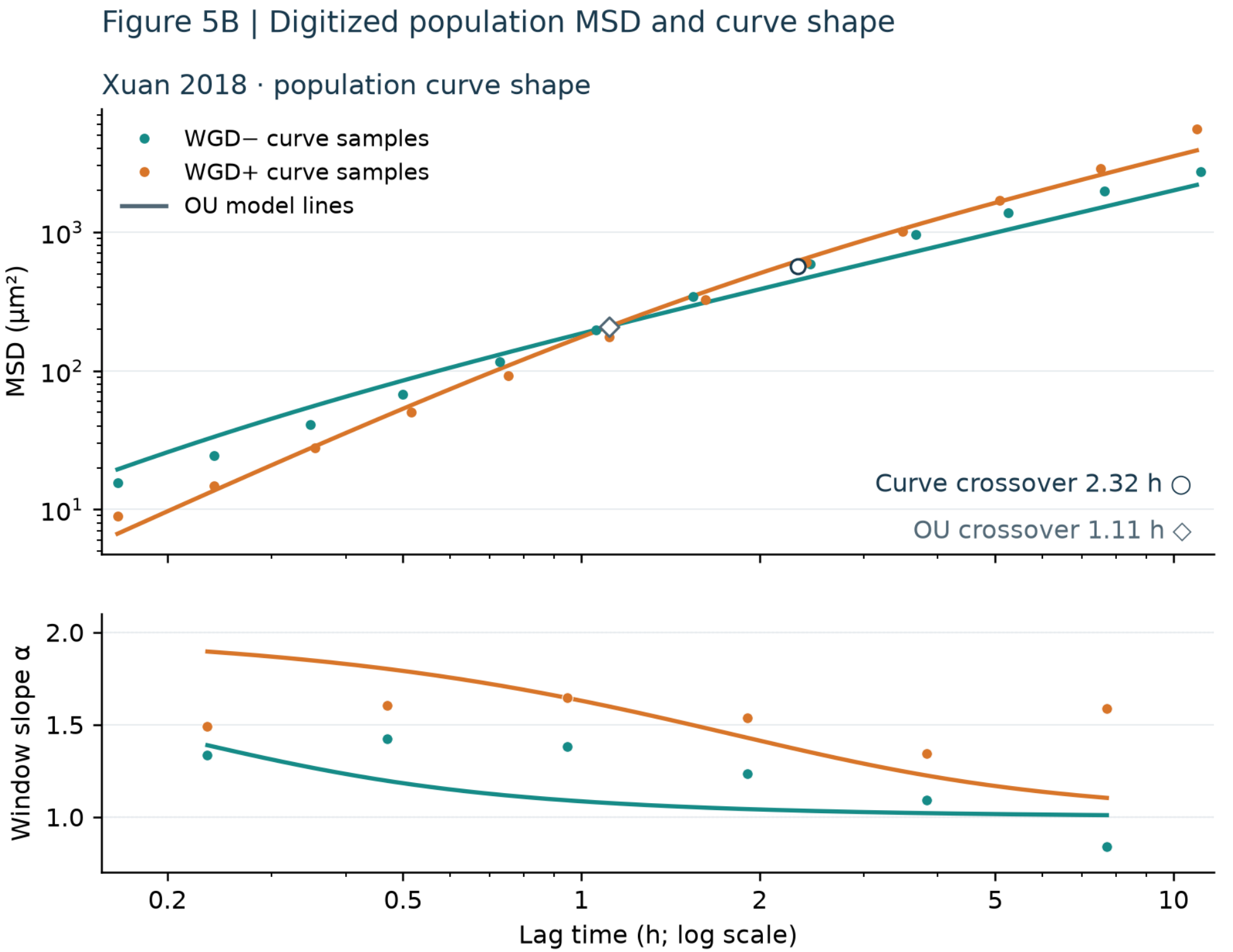


Figure 5B continued: enlarged detail with explicit raster-sample caveat.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation

Existing empirical support, with observed migration summaries kept separate from fitted velocity-memory parameters.

Detailed legend

A, Published/digitized enlarged or polyploid-group to reference-group ratios for nine migration summaries, alongside primary Ornstein–Uhlenbeck (OU) fit predictions. Xuan contributes sampled mean speed and population mean-squared displacement (MSD) at the last shared lag, approximately 10.95 h; Hosny summaries are group medians. Mallin central-bar statistic is not specified by the source and retains the historical mean approximation. These distinct source summaries are not interchangeable measures of persistence.

B, Upper: twelve chosen display samples from each digitized Xuan population MSD curve (small points) and the unchanged fitted analytical OU expectation (solid lines), on logarithmic lag and MSD axes. These are samples of the published raster curves, not recovered experimental lag positions or independent cell observations; the original experimental lag grid is unknown. The open circle identifies the crossover of the full digitized traces at approximately 2.32 h, computed by linear interpolation in native lag and MSD. The open diamond identifies the different fitted OU crossover at approximately 1.11 h. Lower: descriptive slopes of log(MSD) versus log(lag) fitted in six equal log-lag windows of the full digitized curves (points), and matching-window moving OU slopes (solid lines). Each window spans approximately 0.304 decades; both observed and model slopes use the same full window support. Observed raster columns remain correlated and give no biological sample size or confidence interval. These slopes are newly calculated curve-shape summaries, not author-reported local exponents or fitted OU memory times. A constant power-law exponent would appear horizontal; reference values 1 and 2 indicate diffusive and straight-line scaling. The model does not reproduce the complete observed curvature or the observed crossover. The analytical fit, all fitting targets and numerical inputs to A and C–F are unchanged.

C, Representative Xuan primary-fit model-generated tracks, with sampled speed, total path length and directness (net displacement divided by path length). Both groups use identical spatial scales. The tracks are simulations, not published individual-cell trajectories; a slower sampled trajectory can nevertheless have greater endpoint directness. Dashed chords illustrate the contribution of finite-time displacements.

D, With reference-group parameters fixed, vary enlarged-group persistence and velocity-noise ratios. Separate surfaces show change in the descriptive data-misfit score and in the total regularized objective; the selected fit and equal-group reference are marked. The heuristic score is not a calibrated likelihood, and the surfaces are not confidence regions.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

E, Primary paired persistence and velocity-noise estimates for Xuan 2018, Mallin 2023, Jemaà 2017, Jemaà 2023 and the Hosny hypoxic baseline. F, Enlarged/reference persistence folds under the primary 10-minute prior, 5- and 20-minute alternatives and stronger/weaker pooling, with the pooled geometric mean. Priors and pooling define conditional sensitivity scenarios, not uncertainty intervals. The Jemaà comparisons constrain speed alone, leaving persistence/noise tradeoffs dependent on hierarchy and priors.

The fitted population contrasts span stable tetraploids, giant polyploids and treatment-induced polyaneuploid cancer cells (PACCs). The inherited WGD+/WGD− labels in the artwork are comparison shorthand and do not establish matched WGD-only interventions in every study. Hosny observations derive from the recovered preprint; equivalence to the journal version remains unresolved.

Narrative points and conclusions

- Xuan combines lower sampled speed (0.64-fold) with greater late-lag population MSD (2.08-fold); Mallin combines shorter path length (0.52-fold) with greater directness (3.12-fold). Persistence-related behaviour therefore deserves analysis separately from speed.
- Hosny baseline PACCs have greater median path length (1.77-fold) and directness (1.67-fold). Signed migration index is a separate directional summary.
- All five primary OU comparisons infer longer velocity memory. The pooled primary persistence fold is about 15.8, and remains above one across the declared prior/pooling scenarios; this is a conditional model result.
- The fit does not reproduce the complete Xuan curve shape within its raster envelope, and data-only sensitivity has rank 18 of 22. Optimization completion does not establish model adequacy or uniquely identified persistence times.
- The observed Xuan curves cross near 2.32 h while the unchanged fitted OU curves cross near 1.11 h. Matched-window slopes make the remaining curve-shape discrepancy visible; fitted persistence is a conditional summary.
- Lower sampled speed can coexist with greater late-lag displacement in the enlarged-group comparison. Its causal connection to polarity mechanisms requires matched activity and migration measurements.

Proposed measurements and experimental follow-up

The new WGD assay should measure matched trajectories directly, with declared sampling and observation duration, and assess speed, displacement, directness and lag-dependent MSD before considering fitted persistence. An activity-polarity memory

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

measurement would complement these quantities without equating activity and velocity.

Scope, limitations and pending validation

Neither these heterogeneous group comparisons nor the OU fits establish that enlargement alone causes persistence. Cell identity, induction/treatment and acquisition differences require explicit treatment in the new experiments.

The requested observed-point/model-line and curve-shape display is implemented. The original experimental lag grid remains unknown; neither display samples nor raster-derived slopes supply independent biological replicates. The observed/model crossover discrepancy remains visible.

Sources and reproducibility

Refined OU analysis
Original panel legend
Observed folds
Fit results and adequacy
Hosny sensitivity supplement
Revised display, exact preserved inputs and direct-user feedback
Migration display numerical and preservation checks

Figure 6. RESERVED: new WGD motility results

RESERVED MAJOR EXPERIMENTAL RESULTS

New WGD motility assays — all data pending

A Paired biological system

SUM-159 chemically induced WGD+/− pilot

Confirm group identity, size and morphology

PENDING — no observations available

B Imaging and observed tracks

Optimize acquisition and tracking

Declare cadence, duration, eligibility and replicates

PENDING — no observations available

C Measured migration behaviour

Speed, path length, displacement and directness

Lag-dependent MSD under declared observation rules

PENDING — no observations available

D Evaluate the persistence hypothesis

Relate the acquired observations to WGD and size

Retain null or opposite outcomes; specify controls

PENDING — no observations available

Proposed panel plan only. Design and analysis remain to be agreed before interpreting data.

Explicit placeholder. Every panel is pending acquisition, analysis and author review; no mock observations are displayed.

Figure 6. RESERVED: new WGD motility-assay results

A substantial new experimental results section and main figure. No observations are available; the panel layout is a proposal, not an experimental result.

Detailed legend

A, Proposed experimental context: the SUM-159 chemically induced WGD+/− pair is the agreed starting pilot with the Speros collaboration. Imaging should first be optimized for that pair before expansion to additional cell lines. Ploidy/group identity, physical size and morphology need to be documented so the intended comparison is clear.

B, Reserved for representative experimental fields and trajectories under the selected acquisition conditions. The eventual panel should state sampling interval, usable duration, spatial scale, track eligibility and exclusions. Number of independent preparations and biological replicates remains to be specified.

C, Reserved for empirical motility summaries: sampled speed, accumulated path length, endpoint displacement/directness and lag-dependent MSD. The final readouts, follow-up window and analysis plan are to be settled before interpreting the data; this placeholder does not imply that all listed measurements will be technically feasible.

D, Reserved for the measured relationship between WGD-associated enlargement and migration behaviour, including size distributions and controls needed to interpret the group contrast. Any activity/protrusion measurements arising from the mechanism proposals require a separately agreed assay and should be distinguished from the basic motility pilot. Statistical models, uncertainty reporting and sample-size targets remain open.

Narrative points and conclusions

- This major results section will test whether the planned paired experimental system exhibits the migration behaviour motivating the theoretical work.
- Its conclusions must follow the acquired observations, including null or opposing results. Simulation output and literature fits cannot fill this gap.
- Agreement between increased polarity resistance or activity memory and migration persistence would support the proposed connection; causal attribution requires appropriate perturbations and controls.

Proposed measurements and experimental follow-up

Pending decisions include plating density, acquisition duration and cadence, field/condition throughput, tracking and exclusion rules, replicate structure, confirmation of WGD status, size quantification and the statistical analysis. The known throughput versus spatial/temporal resolution trade-off should be resolved by the pilot.

Figure 6. RESERVED: new WGD motility-assay results · continued

Mechanism validation can subsequently examine incumbent extent, early challenger growth, local loss/rebound, site fates and global activity-polarity correlation under characterized perturbations. These activity experiments are proposals alongside the theoretical mechanisms and are separate from the reserved motility-results contribution.

Scope, limitations and pending validation

No pilot acquisition, sample size, effect estimate, significance claim or experimental outcome is supplied by the available evidence.

Sources and reproducibility

Author manuscript vision and supplied collaboration context

Prospective SUM-159 WGD pilot observables

Specified on 23 October 2026, before SUM-159 WGD+/− motility data; no experimental observations were used.

Model drift subset: f ≤ 0.01 and multiple-domain fraction ≤ 0.05. In cells, classify by the observed frequency of coexisting protrusions; model f and native times have no calibrated cellular equivalents.

Cellular observable	Model counterpart	Drift regime: WGD+ / WGD−	Secondary-domain regime: WGD+ / WGD−	Mechanistic interpretation
Common-mode motion of the leading region's two lateral edges	Correlated boundary translation and domain drift (Figs 3D, 4D)	Similar (Holmes/Spring); smaller in Wave-pinning/Goryachev	No common direction identified outside persistent single-domain bouts	Translation; does not identify a Figure 2 route
Secondary-protrusion birth rate per unit perimeter	Persistent no-overlap birth rate (Fig. 4E(i))	Rare in both groups	Model dependent: Spring higher and Goryachev lower at f = 0.1; other trends unresolved	Initiation; model-dependent size response
Takeover incidence within a fixed follow-up window	Five-frame dominance within 20τ after birth (Fig. 4E(ii))	Not applicable when births are too rare	Lower by 20τ; persists through 80τ in the same cohort; lifetime fate unidentified	Incumbent resistance; lifetime success unresolved
Activity/protrusion-heading autocorrelation time	Heading decay and size exponent (Fig. 4B–D)	Longer; approximately size squared only when physical drift is size independent	Size advantage shrinks in the tested models; compare coherence and domain counts	Regime readout; no unique route assignment
Edge displacement after boundary-straddling photoactivation	Weak deterministic edge response (Fig. 3C)	Smaller (Wave-pinning/Goryachev); similar (Holmes/Spring)	Not specified: the pulse assay uses a stable polarized incumbent	Susceptibility within a route, not a unique mechanism

Prospective directions, not SUM-159 observations or a full statistical preregistration. Freeze sampling, segmentation, physical pulse dose and group inference before comparison; model f and native times have no calibrated cellular equivalents.

Birth rates are model dependent: at f = 0.1, bounds identify a Spring increase and a Goryachev decrease with size. Takeover within 20τ is fixed-horizon incidence; its decline persists through 80τ in the same cohort, while lifetime fate and certification selection remain unresolved.

Common translation tests the drift mechanism. Edge pulses refine susceptibility within routes: Wave-pinning and Holmes share an expansion-constrained class in Figure 2 but predict different edge responses. Combine these readouts with Figure 2 first-rise, width and pulse-gap assays to test initiation-limited reset, saturation/expansion and retained progress after visible activity loss. Activity heading, observed migration and OU-fitted memory remain separate readouts.

Prospective pilot observables before SUM-159 data

Specified on 23 October 2026, before SUM-159 WGD+/− motility data; no experimental observations were used.

Detailed legend

Prospective directions, not SUM-159 observations or a full statistical preregistration. Freeze sampling, segmentation, physical pulse dose and group inference before comparison; model f and native times have no calibrated cellular equivalents.

Birth rates are model dependent: at f = 0.1, bounds identify a Spring increase and a Goryachev decrease with size. Takeover within 20τ is fixed-horizon incidence; its decline persists through 80τ in the same cohort, while lifetime fate and certification selection remain unresolved.

Common translation tests the drift mechanism. Edge pulses refine susceptibility within routes: Wave-pinning and Holmes share an expansion-constrained class in Figure 2 but predict different edge responses. Combine these readouts with Figure 2 first-rise, width and pulse-gap assays to test initiation-limited reset, saturation/expansion and retained progress after visible activity loss. Activity heading, observed migration and OU-fitted memory remain separate readouts.

Narrative points and conclusions

- Model drift subset: f ≤ 0.01 and multiple-domain fraction ≤ 0.05. In cells, classify by the observed frequency of coexisting protrusions; model f and native times have no calibrated cellular equivalents.

Proposed measurements and experimental follow-up

Track both lateral half-contrast boundaries of a persistent leading activity/protrusive region. Report their mean-squared physical displacements and cross-correlation over declared lags, including failed single-domain eligibility and changes in region identity. Compare physical displacement before normalizing to perimeter. The two edges are strongly correlated in model exemplars, so report their covariance rather than assuming independence. Report common-mode displacement (left plus right)/2 separately from width fluctuations (right minus left); a single-edge MSD alone cannot distinguish these modes.

With full images retained, call a new connected protrusive/activity region a birth only if it overlaps no region in the preceding frame and persists for at least five consecutive saved frames. Divide by observed perimeter-time exposure; count splits separately. Freeze segmentation, cadence and overlap rules before group comparison.

For each qualifying birth, measure shortest centroid arc distance from the incumbent minus the incumbent half-width, estimated as excess amount divided by twice the declared plateau height. A takeover requires five consecutive frames of dominance.

Prospective pilot observables before SUM-159 data · continued

Report right censoring and a common follow-up horizon; distinguish all births from any subset whose identity is certifiable. The revised model comparison uses a common 20-noise-correlation-time follow-up and pools $f > 0.01$; these conditioning choices must be separated from the cellular analysis. This is fixed-horizon incidence, not a lifetime takeover probability. Record time since birth, cumulative incidence and right censoring. In the same model cohort with at least 80τ follow-up, the decline persists from 20τ to 80τ; finite observation and certification selection still prevent a population lifetime claim.

Define the activity/protrusion vector from the full spatial signal; estimate heading autocorrelation at declared lags with a fixed coherence rule and report excluded pairs. Compare groups at matched analysis lags and observation duration. Keep cell displacement/velocity persistence as a separate observed readout.

Apply a weak, brief photoactivation spot centred on a lateral leading-edge boundary, matched in physical width and duration across ploidy groups. Track both edges through relaxation, subtract a no-pulse control, and check half/double amplitudes. A response supports a mechanism only to the extent that its size trend differs among model predictions. At model $f = 0.001$, largest/smallest pulse-response gains are 0.701 (Wave-pinning), 0.914 (Holmes), 0.989 (Spring), and 0.0878 (Goryachev); these are native-model size contrasts, not numerical cellular fold predictions. Separately repeat the Figure 3A–B assay with cue placements straddling and separated from the incumbent boundary, measuring half-turn time to distinguish sliding from takeover; the weak local pulse and the finite turning cue are distinct assays.

Scope, limitations and pending validation

No SUM-159 motility data are available in this revision. Directions are prospective and must not be described as confirmed experiments.

Native model times and f are not calibrated to cells. Event holds measured in frames depend on acquisition cadence; select and report that cadence before comparing experimental groups.

The compact model records do not identify every birth, split or local plateau height. Model bounds and certified-subset probabilities do not establish the unconditional cellular birth/takeover directions.

The new observables do not uniquely distinguish the Figure 2 response classes. Weak boundary susceptibility and finite-cue sliding/takeover are different properties; the within-20τ size contrast cannot distinguish slower completion from a change in eventual success.

Sources and reproducibility

Prospective directions and operational definitions

Supplementary material and supporting analyses

Figure 3: edge-response / secondary-domain revision supplement

New measurements, identification limits, numerical checks and preceding supplementary evidence.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round5_20261023/figure3/figure3_round5_review_supplement.pdf

Physical diffusion prediction and observed drift

Size-normalized prediction and observed drift

Normalized integrated boundary response

Methods and covariance derivation

Both boundary assays and numerical checks

Reviewed main/supplement preservation

Figure 4: edge-response / secondary-domain revision supplement

New measurements, identification limits, numerical checks and preceding supplementary evidence.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round5_20261023/figure4/figure4_supplement.pdf

Round5 complete campaign analysis

Scientific identity/enclosure validation

Birth and heading-loss conditions

Takeovers with common follow-up

Original campaign manifest

Native-amplitude heading accounting

Native-amplitude condition metrics

Figure 4: corrected event scoring and full closure comparisons

Scorable-event coverage, missing-angle bounds, both detector-free cadences and all closure failures. Preceding editions remain linked below.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round4_20261023/figure4/figure4_supplement.pdf

Scientific methods and regime-specific counts

Figure 4: round-3 analysis and null comparisons

Pointwise and smooth size exponents, drift/jump cross-checks, sensitivity, valid fractions, replication and analytic null checks. Missing estimates and finite-rate precision shortfalls are identified in the scientific readme.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round3_20261022/figure4/figure4_supplement.pdf

Unexpected results, data status and methods

Supplementary material and supporting analyses · continued

Figure 2: round-3 evidence and preserved prior panels

The complete supplement adds the per-width replacement curves and the deBelly native endpoint audit, then preserves the complete round-2 main and every earlier supplement page. Superseded fate maps, ratios and class tables are historical evidence.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round3_20261022/figure2/figure2_round3_supplement.pdf

Round-3 reader and source data

Exact preservation receipt

Figure 3: round-3 evidence and every preserved round-2 panel

The supplement retains 90/99%-heading time curves, the preceding four-placement round-3 main and legend including C = 47.6 profiles, and the complete round-2 main and supplement. Original source files remain unchanged.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round3_20261022/figure3_single_C/figure3_round3_supplement.pdf

Round-3 reader and native source coordinates

Exact supplement preservation

Historical Figure 4: previous supplement and decision record

Historical evidence from the preceding Figure 4 edition, preserved unchanged. The comprehensive supplement retains the full designed condition grid, unavailable measurements, finite-horizon bounds, native histories, controls and the stated numerical and observational sensitivities. Earlier figure versions remain development history. S0–S12 statistical core and complete native-atlas appendix index; native pages are in all linked appendix volumes. Every numbered appendix below is part of the complete supplement.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/signature_figure4_20261020/supporting/vectors/figure4_revision_complete_supplement.pdf

Figure reader and source data

Independent completion review

S11 native atlas appendix 001

S11 native atlas appendix 002

S11 native atlas appendix 003

S11 native atlas appendix 004

S11 native atlas appendix 005

S11 native atlas appendix 006

S11 native atlas appendix 007

S11 native atlas appendix 008

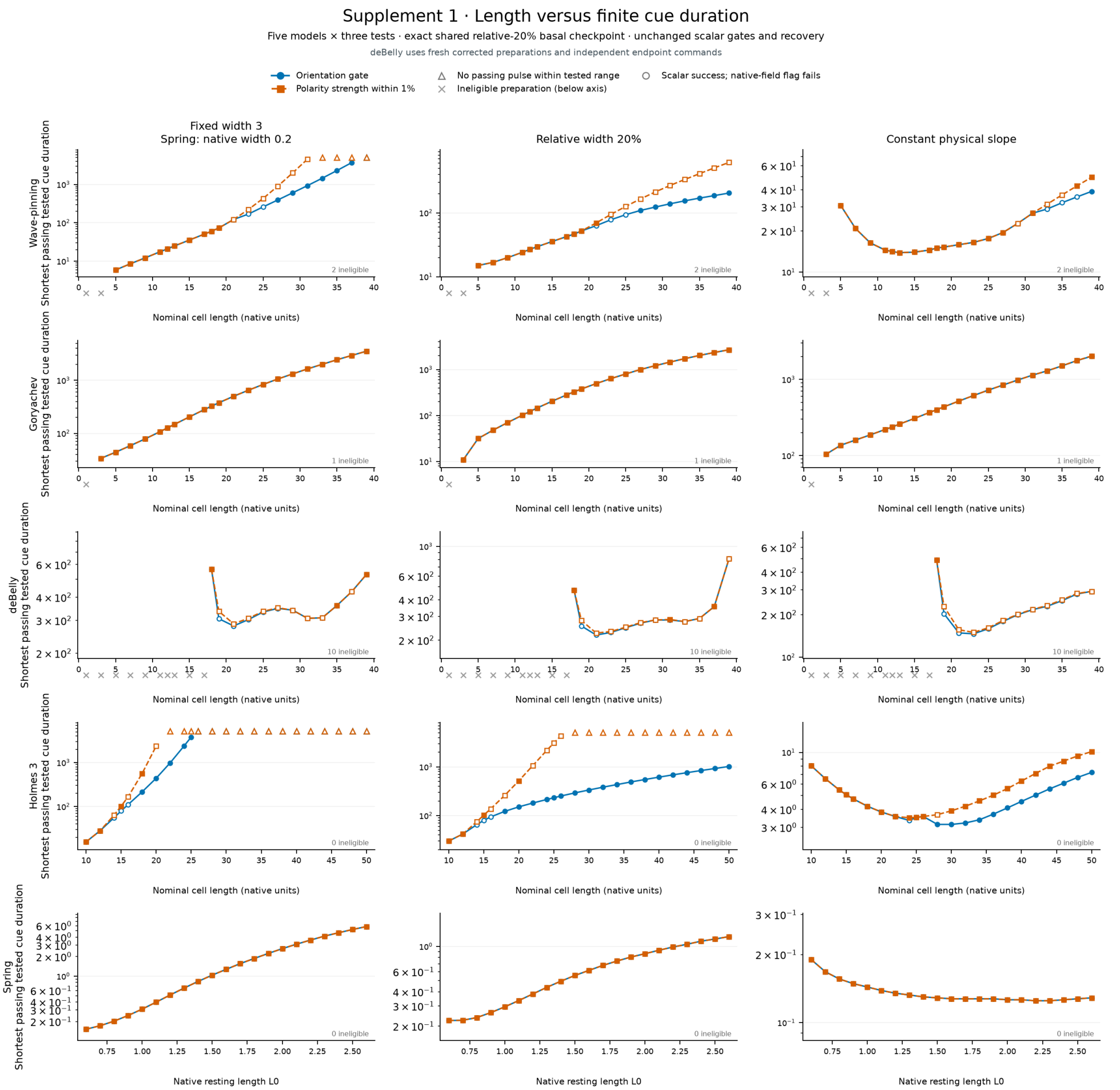
S11 native atlas appendix 009

S11 native atlas appendix 010

Supplementary material and supporting analyses · continued

S11 native atlas appendix 011
S11 native atlas appendix 012
S11 native atlas appendix 013

S1. Full cue-duration response curves



Both patches use a constant peak across all lengths; Goryachev K=64, peak=6.4. Gradients retain one constant physical slope per model.
Each positive duration restarts the original checkpoint. Full strength adds the 1% reversed initial-polarity band to the orientation gate.
Open markers retain failing field-tail flags. Censoring marks explicit tested limits; missing endpoints break lines. Native times and scales differ.
Data generated 2026-10-08T19:26:33.200718+00:00; producer finished. Every imported and fresh trial passed independent native-array audit.

Canonical corrected five-model duration curves; source working number Supplement 1.

Earlier supporting material: S1. Full cue-duration response curves

Essential curve context for the main comparison heatmap.

Detailed legend

Five model rows and three finite-cue columns retain the complete sampled length dependence. Coloured curves distinguish orientation from strength replacement. Open success markers retain scalar successes that fail the separate native-field tail flag; triangles indicate no passing pulse within the tested range; crosses indicate ineligible preparations. Missing values are not zero response times. Each duration excludes preparation and the subsequent 10,000-unit basal recovery. Spring uses native length and time. See Figure 1 for exact gates and stimulus conventions.

Narrative points and conclusions

- Corrected deBelly is nonmonotone across all three geometries. The positive rank coefficients summarize association rather than its full curve shape.
- Constant-slope Spring shows the opposite association to its patch responses. Holmes orientation under that cue varies with little monotone association. Censoring and native-field flags restrict interpretation even where rank correlation is strong.

Sources and reproducibility

Current figure legend and gate definitions

Supplementary material and supporting analyses

Earlier supporting material: S2. Numerical qualification and preparation provenance

Exact common starting states, duration-boundary audits, corrected deBelly endpoint controls and selected mesh diagnostics; source histories remain distinguishable from compact committed summaries.

polarity1d/manuscript/THREE_TESTS.md

Earlier supporting material: S3. Mechanism and stochastic sensitivity

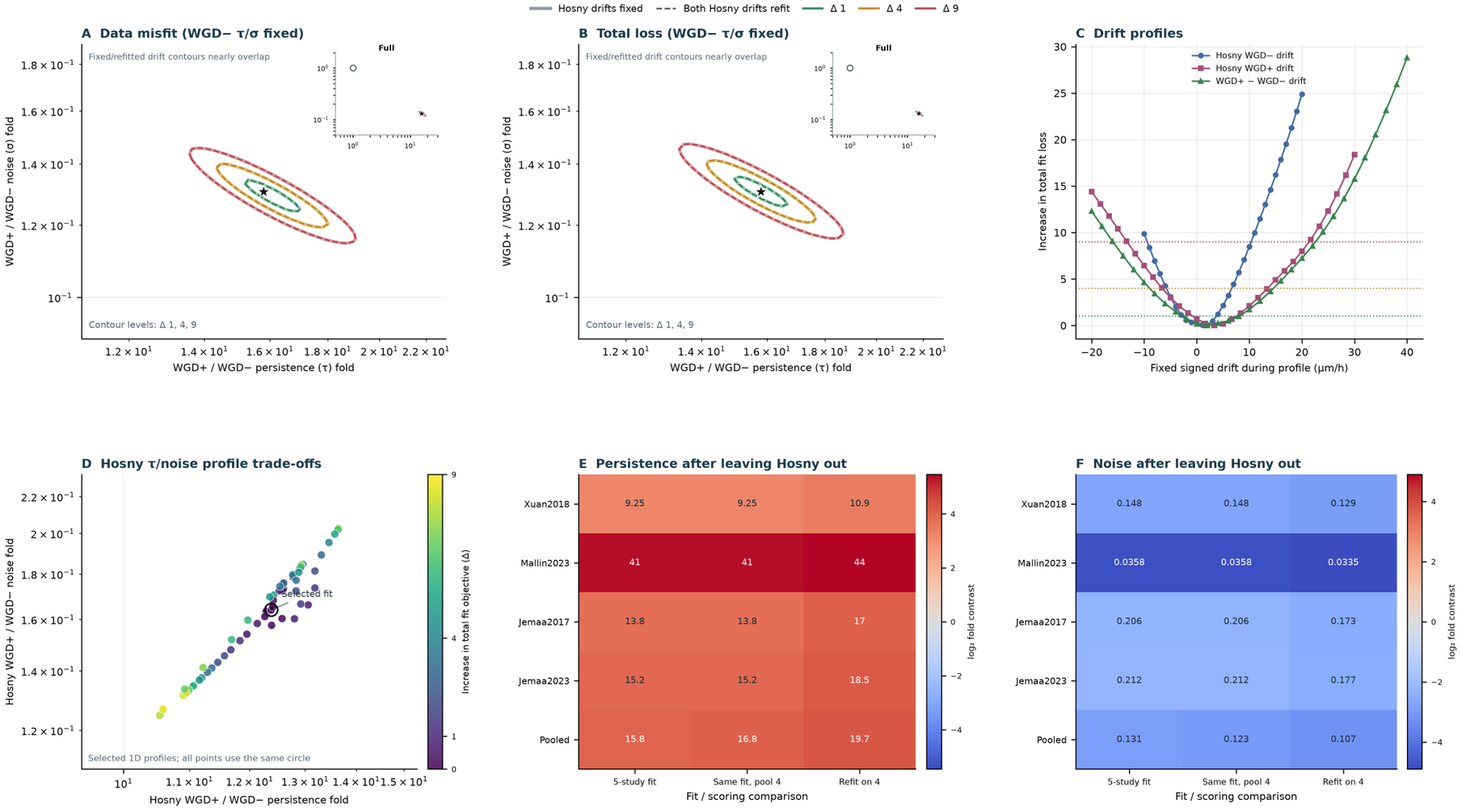
Complete gain/recovery grids, clock qualification, finite-cue definitions, outcome tables, site/vector diagnostics and numerical/noise controls accompany the mechanism checkpoint. The historical Figure 4 edition also displays both strengths, complete half-window diagnostics, physical-site eligibility and matched C20/C60 native examples; compact source tables support portable rerendering.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/VISION_CHECKPOINT_REQUIREMENTS.md

S4. Migration-model adequacy and Hosny sensitivity

How much does the Hosny drift assumption change the fitted contrast?

Primary Gaussian OU hierarchy · profile panels reoptimize remaining parameters · Δ values are descriptive fit sensitivity



Objective is arbitrarily weighted and has no calibrated statistical or biological units; it is not a likelihood. Profile ranges are unions of sampled 1D fits, not joint uncertainty regions. Hosny contributes medians to fitting.

Existing Hosny sensitivity supplement from the refined OU analysis.

Earlier supporting material: S4. Migration-model adequacy and Hosny sensitivity

Supporting conditional sensitivity analysis; it is not an independent biological replication.

Detailed legend

The retained supplement explores signed-drift and leave-Hosny-out sensitivity within the declared OU hierarchy. Its fitted changes and descriptive-score profiles are conditional on the model and supplied summaries, not confidence intervals. Data-only identifiability, Xuan population-curve residuals and prior/pooling assumptions remain part of the adequacy assessment. Representative movie diagnostics are separate: acquisition timing is uncertain and nuclear appearance does not establish ploidy.

Sources and reproducibility

Original Hosny supplement caption

Sensitivity methods and results

Earlier supporting material: Gain and small-loss recovery: supporting evidence

Detailed legend

Calculated native input gain is separate from observed post-cue activity. The small-loss recovery contrast persists through stored finite trajectories but weakens with time; actual damage is only about0.22–0.285% of core activity, with no reporter calibration.

Scope, limitations and pending validation

Small-loss assays are not the finite optical challenge or a universal stability measure.

Sources and reproducibility

Finite-time and amplitude audit

All97 gain/recovery values

Earlier supporting material: Selected native pulse and causal controls

Detailed legend

WP L15 T20/T60 starts from the same basal state and illustrates withdrawal-dependent domain survival. Evolved interventions have endpoint- and cue-specific limits.

Scope, limitations and pending validation

WP4× mobility passes orientation only at one dose; both conditions fail the stronger full-polarity-strength gate. Mobility is changed during cue and recovery.

Sources and reproducibility

Retained WP short/long pulse artwork

Endpoint-qualified controls

Supplementary material and supporting analyses

Earlier supporting material: Stationary-cue endpoint-vector alternative

An alternative view displays all 240 measured endpoint vectors on equal coordinate scales, with weak-resultant, overlap and finite-tail flags retained. It is available for discussion without replacing the complete withdrawal/retention map in the current review.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/standard_angles_sizes/vector_grid/README.md

Earlier supporting material: Spring: every native medoid

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_all_histories.png

Earlier supporting material: Spring magnitude, geometry and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_magnitude_geometry_controls.png

Earlier supporting material: Spring full/half-window memory

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_memory_windows.png

Earlier supporting material: Spring increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_increment_tails.png

Earlier supporting material: Spring short/long-lag geometry and circular mixing

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_endpoint_lag_sensitivity.png

Earlier supporting material: Spring largest sampled increment native context

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_large_increment_context.png

Earlier supporting material: All chemical models including Goryachev

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_stochastic_revision.png

Earlier supporting material: Chemical memory: both strengths

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_both_strengths_memory.png

Supplementary material and supporting analyses · continued

Earlier supporting material: Chemical magnitude and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_magnitude_controls.png

Earlier supporting material: Chemical increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_increment_tails.png

Earlier supporting material: Chemical site-gate sensitivity

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_site_gate_sensitivity.png

Earlier supporting material: Original chemical medoids, all18 conditions

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_selected_histories.png

Earlier supporting material: Goryachev lowest-resultant diagnostic

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_lowest_resultant_diagnostic.png

Earlier supporting material: Spring active amount and spatial first harmonic

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_amount_spatial_balance.png

Earlier supporting material: Preserved complete mechanism and assay checkpoint

The preceding outline retains all-five mechanism-to-measurement sheets, ring gain/recovery and finite-cue maps, and noise magnitude, controls and full/first/last-window diagnostics. Its numerical evidence is preserved; the revised main-figure claims are stated in this edition.

manuscript_outline/outline_v2.pdf

Prior all-five mechanism evidence crosswalk

Prior complete ring and noise checkpoint

Prior physical motion, magnitude and coverage evidence

References and evidence provenance

Reviewed primary literature and extraction map
Scientific scope and author direction
Direct-user preference evidence and foundational figure audit
Mechanism checkpoint intake and live-producer evidence
Source checkout, evidence selection and reproduction
Independent critical assessment and repair decisions
Portable original/renumbered mechanism replay and source closure
Second-edition integration decisions and validation scope
Figures 2–4 revision specification
Figure 2 October 20 revision brief
Figure 3 October 20 revision brief
Figure 4 October 20 revision brief